

Automatic ISUP Grade Classification in Prostate Histopathological Images Using EfficientNet and Ordinal Focal Loss

Firstname Lastname ¹, Firstname Lastname ² and Firstname Lastname ^{2,*}

¹ Affiliation 1; e-mail@e-mail.com

² Affiliation 2; e-mail@e-mail.com

* Correspondence: e-mail@e-mail.com; Tel.: (optional; include country code; if there are multiple corresponding authors, add author initials) +xx-xxxx-xxx-xxxx (F.L.)

Abstract

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Keywords: keyword 1; keyword 2; keyword 3 (List three to ten pertinent keywords specific to the article; yet reasonably common within the subject discipline.)

1. Introduction

Prostate cancer is one of the most common malignant neoplasms in men, representing a significant challenge for public health [1]. Statistics reported by the International Agency for Research on Cancer indicate that, in 2022, this neoplasm was the fourth most common worldwide, accounting for 7.3% of all cancer cases [2]. In this context, tumor grading plays a central role in defining clinical management and prognosis. The Gleason system, based on the sum of the two predominant histological patterns, evolved into the grading system proposed by the International Society of Urological Pathology (ISUP), organized into five groups ranging from 1 to 5. This system provides improved prognostic stratification and is widely used in clinical practice [3]. Traditional histopathological evaluation of Whole-Slide Images (WSIs) relies on manual analysis performed by pathologists. Given the large number of fields that must be examined on a single slide, the evaluation process becomes a repetitive task that requires long hours, sustained focus, and professional expertise. In this context, fatigue resulting from prolonged analysis may introduce inter- and intra-observer variability in the results, which directly influences patient clinical management [4]. Consequently, in recent years, there has been a significant growth in the use of Artificial Intelligence (AI) techniques to automate the analysis of histopathological examinations and assist in prostate cancer diagnosis, aiming to mitigate the issues mentioned above [5]. In this scenario, Convolutional Neural Networks (CNNs) have represented the state of the art in medical image analysis [6], being capable of extracting relevant features and combining

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complex visual patterns to identify specific pathological conditions [7]. This capability makes this approach particularly promising for the automated analysis of Gleason patterns and ISUP grade groups. Thus, this approach contributes to the standardization of prostate cancer diagnosis and to the reduction of intra- and inter-observer variability inherent to conventional histopathological evaluation. In addition to reducing diagnosis time, such an approach may enable more precise and reproducible clinical management. This work proposes a structured pipeline for the automatic classification of ISUP grade groups in prostate histopathological images. The main contributions are: (i) a hybrid loss function that combines ordinal penalization and focal loss, considering the hierarchical nature of the problem and the class imbalance in the dataset; (ii) an automated noise removal method based on uncertainty estimated from the average predictions of EfficientNet models (B0, B1, B2, and B3); and (iii) a comparative evaluation of the EfficientNet B0, B3, and B7 architectures.

2. Materials and Methods

This section presents the methodological procedures used to develop and evaluate the proposed approach. The method, illustrated in Figure 1, was structured into three main stages: (i) materials, (ii) preprocessing, and (iii) feature extraction and model training. Each stage is described in detail in the following subsections.

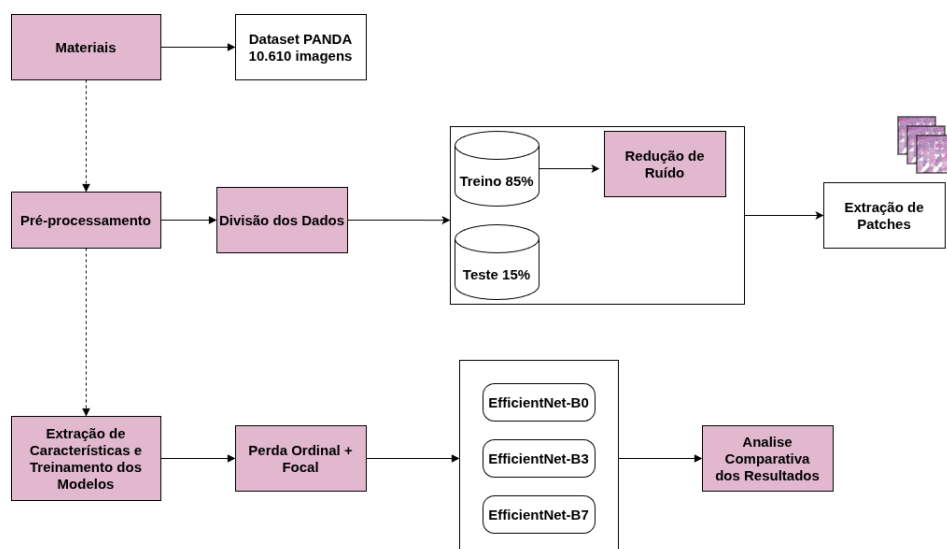


Figure 1. Proposed Method

2.1. Materials

In this work, the PANDA dataset (*Prostate Cancer Grade Assessment*) was used, comprising approximately 10,616 complete histopathological images (*Whole Slide Images - WSIs*) from prostate biopsies [8]. This dataset was released as part of the PANDA challenge on the Kaggle platform and has been widely used in research on the automatic classification of prostate cancer. The images in the dataset are annotated with the Gleason score and Gleason grade, providing clinical labels used for disease grading. In this study, only the public version of the dataset was used, without access to the private test set provided during the challenge.

The data were collected from two distinct medical centers, *Radboud University Medical Center* and *Karolinska Institute*, introducing institutional variability into the images. This variability is associated with differences in slide preparation protocols, scanning procedures,

and annotation criteria, thereby making the experimental scenario more closely resemble real clinical practice.

In addition, the dataset contains noise inherent to the slide acquisition and annotation processes. Among these, visual artifacts include pen marks made by pathologists during slide analysis, as illustrated in Figure 2. There are also labeling-related sources of noise, such as disagreement among annotators in assigning histopathological grades. These factors increase the complexity of the automatic classification task and reflect challenges present in digital pathology practice.



Figure 2. Example of a visual artifact present in a WSI from the PANDA dataset. On the left, the complete slide is shown with the region of interest highlighted in red. On the right, an enlarged view of the selected region reveals a pen mark over the prostate tissue.

Another relevant aspect of the dataset is the unequal distribution of samples among the six ISUP grading classes (ISUP 0 to ISUP 5). It is noteworthy that ISUP grade 0 indicates the absence of malignant lesions. A higher concentration of samples is observed in ISUP classes 0 and 1, while the remaining classes are less represented. The complete distribution of samples is presented in Figure 3.

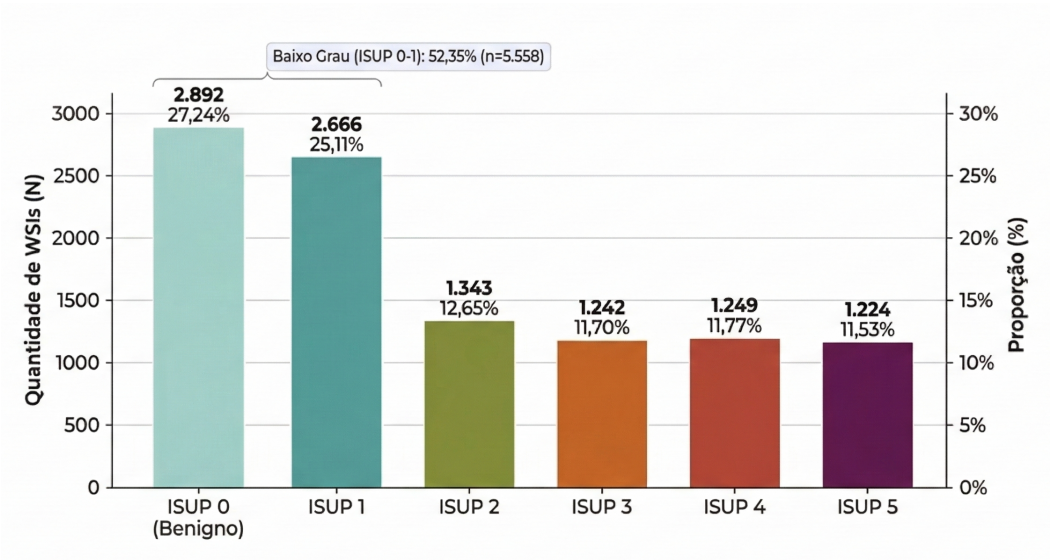


Figure 3. Distribution of samples in the PANDA dataset

2.2. Preprocessing

During the preprocessing phase, the data were initially divided into training and testing sets to ensure a clear separation between model fitting and evaluation. Subsequently, a noise reduction technique based on entropy was applied, followed by patch extraction based on the information density of each slide.

For the noise-reduction stage, four architectures based on the *EfficientNet* backbone (B0, B1, B2, and B3) were used to compute Shannon entropy to quantify prediction uncertainty [9]. The average entropy of the four models was calculated for each sample in the training set, and the top 10% percentile was selected, forming a dataset of potentially noisy samples. From this subset, four scenarios were evaluated: retaining all data and removing 100%, 50%, and 20% of the images identified as noisy. The best performance was obtained when removing 20% of these images (180 samples), indicating that many samples, despite presenting high uncertainty, still contribute relevant information to the problem within the dataset context.

In the final stage of preprocessing, new images were generated from the PANDA WSIs through patch extraction, selecting regions containing relevant tissue. This approach was necessary because histological material occupies only part of the slide; without patch extraction, processing the original WSIs directly would incur high computational cost and analyze large non-tissue areas, as illustrated in Figure 4.

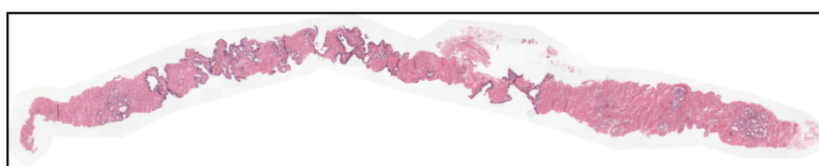


Figure 4. Example image from the PANDA dataset

To determine the best extraction strategy, the reorganization of images into *patches* was evaluated using configurations of 5×5 (25 *patches*), 6×6 (36 *patches*), and 7×7 (49 *patches*). As illustrated in Figure 5, the 5×5 configuration presents a higher density of tissue and a smaller proportion of empty areas; however, the reduced number of fragments may lead to the loss of potentially relevant regions, impairing the representation of more subtle lesion patterns. On the other hand, the 7×7 configuration preserves greater spatial coverage of the slide but includes many regions without tissue and produces high-dimensional images (1792×1792), increasing memory requirements and processing time. Therefore, the intermediate configuration of 36 *patches* was selected, as it preserves most of the tissue information while maintaining relatively reduced dimensionality (1536×1536), balancing quality, coverage, and computational efficiency.

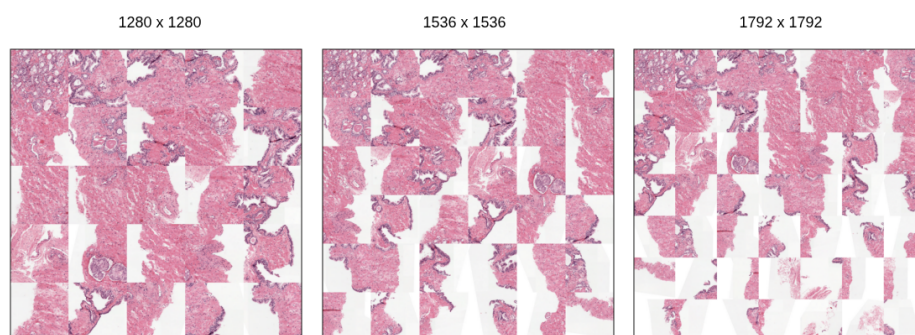


Figure 5. Illustration of patch extraction with different grid sizes: 5×5 (left), 6×6 (center), and 7×7 (right).

2.3. Feature Extraction and Model Training

This section presents the proposed approach for training the individual classifiers, including the definition of hyperparameters, regularization strategies, and the loss function. After preprocessing, model training and evaluation were initiated. *EfficientNet*-B0 was used as the base model for hyperparameter tuning. Once the configuration was defined,

the experiments were extended to EfficientNet-B3 and B7, enabling comparisons across different complexity levels within the same family and analysis of the impact of increased representational capacity on performance and generalization.

Initially, *Binary Cross-Entropy* (BCE) was adopted as the baseline loss function due to its widespread use in multilabel classification scenarios. From this initial configuration, exploratory experiments were conducted to assess training stability and the impact of hyperparameters on model generalization. Variations in the learning rate, the number of layers unfrozen during *fine-tuning*, and regularization strategies through *Dropout* were evaluated. To select the best hyperparameter combination, a systematic grid search was performed using the EfficientNet-B0 model. This search was conducted on a subset corresponding to 20% of the training set, reserved for internal validation, enabling identification of the best-performing configuration before the final training stage.

Despite evaluating these configurations during training, performance stagnation was observed, which could be attributed not only to the choice of hyperparameters but also to the formulation of the loss function. Given that prostate cancer grading is ordinal, with errors between adjacent classes less severe than those between more distant classes, the problem was reformulated as an ordinal regression task.

In this formulation, ISUP grade prediction is not treated as a multiclass problem with independent categories but rather as a sequence of cumulative binary classifications. For an ISUP grade $k \in \{0, \dots, 5\}$, the label is encoded as a cumulative binary vector in which the first k elements assume value 1 and the remaining elements 0; for example, grade 3 is represented as $[1, 1, 1, 0, 0]$. The ordinal loss is then applied independently to each threshold in the vector, yielding a total error that corresponds to the sum of cumulative discrepancies. This modeling imposes a penalty proportional to the distance between classes, such that more distant errors incur greater cost, preserving the hierarchical structure of the problem.

To mitigate class imbalance (Section 2.1), the ordinal loss was combined with *Focal Loss* [10–12]. This function modulates cross-entropy to reduce the influence of highly confident predictions, prioritizing samples with greater diagnostic uncertainty. In this study, the focusing factor $\gamma = 2.0$ was used to attenuate the contribution of easy examples, and the weight $\alpha = 0.25$ was used to balance the relative importance among classes. This strategy directs optimization toward more complex and underrepresented cases, which are essential for precise discrimination between ISUP grades.

2.4. Evaluation Metrics

The metrics used were **accuracy** and κ_{quad} (Quadratic Weighted Kappa). Accuracy measures the proportion of correct predictions relative to the total number of evaluated instances [13]. The κ_{quad} metric quantifies the degree of agreement between predicted and true classes while incorporating penalties proportional to the magnitude of the error, making it particularly suitable for ordinal problems [14].

The variability of the metrics and their respective confidence intervals were estimated using nonparametric bootstrap [15]. Based on predictions from the test set, 1,000 resamples with replacement were generated while preserving the original dataset's size. For each iteration, the performance indicators were recalculated, enabling estimation of the mean, standard deviation, and 90% confidence interval (CI) based on the 5% and 95% percentiles. This approach provides a robust uncertainty estimate without requiring parametric assumptions about the data distribution.

3. Results

This section presents the implementation environment, the selection of the best configuration pipeline, and the impact of the hybrid Ordinal-Focal loss function.

3.1. Implementation Environment

The experiments were conducted on a Linux workstation equipped with an RTX 3060 GPU, an AMD Ryzen 5 5600X CPU, and 32 GB of DDR4 RAM. In addition, PyTorch, CUDA, and OpenSlide were used to read WSIs and extract patches.

3.2. Initial Stage - Hyperparameter Configuration

The initial training stage, as described in Section ??, enabled the identification of the optimal hyperparameter configuration for the model. The evaluated values and the selected configuration are presented in Table 1.

Parameter	Evaluated values	Selected value
Learning rate	3×10^{-4} , 3×10^{-3} , 1×10^{-4}	3×10^{-4}
Number of epochs	50	50
Unfrozen layers (fine-tuning)	100, 150	150
Dropout	0.4, 0.5, 0.6	0.4
Batch size	2	2
Loss function	BCE	BCE
Optimizer	Adam	Adam

Table 1. Evaluated hyperparameters and the configuration selected after grid search.

3.3. Impact of Noise Removal and Loss Formulations

Table 2 presents the incremental performance evolution of the *EfficientNet-B0*. The strategies were applied cumulatively, so that each configuration incorporates the improvements from the previous stage.

Initially, noise removal was applied to the *baseline* model, increasing the κ_{quad} from 0.826 to 0.833. This result indicates that correcting dataset inconsistencies contributed to greater training stability. Subsequently, while maintaining noise removal, replacing BCE with the ordinal loss increased the value to 0.851. This improvement demonstrates that explicitly modeling the ordinal relationship among classes improves predictive performance. This configuration also presented the lowest standard deviation (0.009), indicating greater stability.

Finally, the hybrid strategy was applied to the ordinal formulation, resulting in the best observed performance ($\kappa_{\text{quad}} = 0.856$). This combination allowed the simultaneous treatment of the ordinal structure of the classes and the dataset imbalance. Overall, the sequence of refinements yielded an absolute gain of 3% relative to the *baseline* model.

Configuration	Quadratic Kappa			Accuracy (%)		
	Result	Std	CI 95%	Result	Std	CI 95%
BCE (<i>baseline</i>)	0.826	0.012	[0.806; 0.846]	0.592	0.012	[0.572; 0.612]
BCE + Noise Removal	0.833	0.013	[0.812; 0.853]	0.638	0.116	[0.619; 0.657]
Ordinal Loss	0.851	0.009	[0.833; 0.869]	0.608	0.126	[0.587; 0.628]
Ordinal + Focal (Hybrid)	0.856	0.011	[0.833; 0.876]	0.669	0.011	[0.635; 0.683]

Table 2. Impact of the loss function formulation on EfficientNet-B0 performance

The incorporation of the ordinal focal loss into the training process improved model performance. Although the kappa coefficient increased modestly (approximately 0.5%), the joint analysis with accuracy reveals a gain of more than 6 percentage points. This increase indicates that the model trained with focal loss was more precise in exact predictions.

This conclusion is reinforced by the analysis of the confusion matrices shown in Figure 6. The model without focal loss (Figure 6(a)) presents lower values along the main

diagonal, especially for the ISUP 3 class, whose accuracy was 0.365, with considerable dispersion toward ISUP 2 (0.303) and ISUP 4 (0.212).

In contrast, the model with ordinal focal loss (Figure 6(b)) shows consistent improvements across several classes, including ISUP 0 (0.857 vs. 0.711), ISUP 3 (0.492 vs. 0.365), and ISUP 4 (0.652 vs. 0.498), demonstrating greater discriminative capability, including in intermediate classes that are traditionally more difficult to classify.

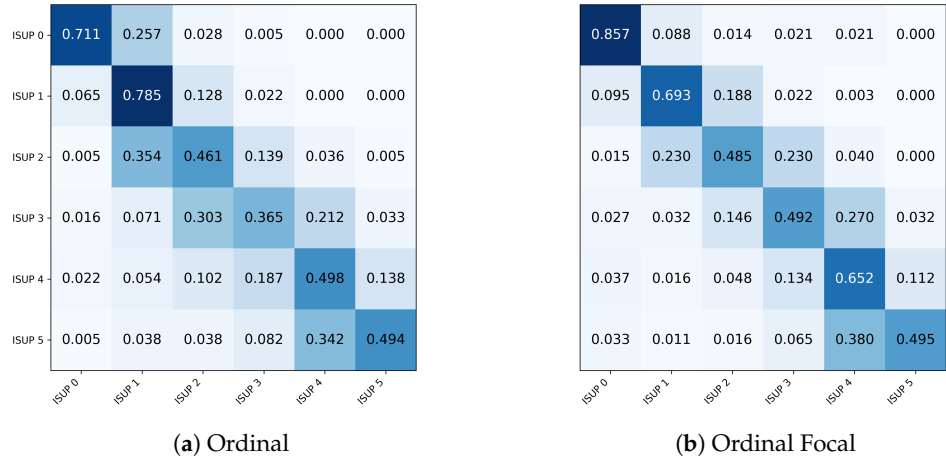


Figure 6. Normalized confusion matrices of models trained with ordinal loss (left) and ordinal-focal loss (right).

3.4. Comparison Between EfficientNet Architectures

After defining the optimal hyperparameters and loss function using EfficientNet-B0, the same experimental configuration was applied to EfficientNet-B3 and EfficientNet-B7, representing intermediate- and high-complexity models, respectively. The goal was to evaluate whether the methodological refinement developed for the base model would produce additional gains when applied to architectures with greater representational capacity within the same family.

The results are presented in Table ?? . It can be observed that EfficientNet-B0 achieved the best performance in both quadratic kappa and accuracy. The deeper architectures (B3 and B7) did not show statistically significant improvements and even exhibited a slight degradation in the mean values of the evaluated metrics.

This behavior may be associated with dataset size or with the greater tendency toward overfitting in models with more parameters. Therefore, EfficientNet-B0 demonstrated the best balance between predictive performance, statistical stability, and computational cost.

4. Discussion

Based on the conducted experiments, the hybrid loss function demonstrated a direct impact on training robustness, increasing the baseline accuracy from 0.592 to 0.669 (+7 percentage points) and the κ_{quad} from 0.826 to 0.856 (+3 percentage points), as shown in Table 2.

The comparative analysis with previous studies (Table ??) shows that although some works report high absolute metrics, they often do so in simplified scenarios, with less challenging validation settings or isolated metrics.

For example, [16] obtained $\kappa = 0.931$ in internal validation but 0.801 in external validation. In contrast, our model achieved $\kappa = 0.856$ directly on the noisy public PANDA dataset, indicating stronger practical robustness. Similarly, [17] reported an accuracy of 91.80% but $\kappa = 0.608$, suggesting relevant ordinal inconsistencies.

Unlike these approaches, the proposed model performs full grading across all six ISUP levels (0–5), explicitly incorporates the ordinal nature of the disease through a hierarchy-sensitive loss function, and reduces extreme classification errors, thereby providing a better balance among statistical performance, clinical coherence, and generalization.

5. Conclusion

In this study, a methodology was developed to automatically classify ISUP grades in prostate histopathological images, addressing relevant challenges such as class imbalance, label noise, and the ordinal nature of tumor progression. The introduction of a hybrid loss function, combining ordinal regression and focal loss, improved model performance by explicitly capturing the hierarchical relationship among malignancy grades. Notably, the largest performance gain was achieved through the application of the ordinal loss, which, when combined with noise removal, increased the κ_{quad} index by approximately 2.5%, while maintaining high stability (standard deviation of 0.009). Overall, the proposed approach improved the κ_{quad} index from 0.826 to 0.857 (an increase of approximately 3%) and the accuracy from 0.592 to 0.669 (an increase of about 7%) compared to the *baseline*.

In comparing architectures within the EfficientNet family, we observed that increasing model complexity did not improve performance, with a slight degradation even in the deeper variants. These results indicate that more complex architectures require more refined training adjustments to reach their full potential. Despite the strong results, the main limitation of this study is the use of a single, albeit multicenter and heterogeneous, dataset, highlighting the need for future validation on external datasets to confirm the robustness and generalizability of the proposed method.

As future work, we propose investigating more recent architectures based on *Trans-formers*, particularly *Vision Transformers* (ViT) and hybrid *CNN-Transformer* models, which may better capture global dependencies in histopathological images. Additionally, *ensemble learning* techniques may be explored to improve the model’s robustness and generalization. These approaches have the potential to further improve system performance, particularly in classifying intermediate ISUP grades, thereby expanding their applicability as diagnostic decision-support tools across diverse clinical contexts.

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Abbreviations

The following abbreviations are used in this manuscript:

MDPI	Multidisciplinary Digital Publishing Institute
DOAJ	Directory of open access journals
TLA	Three letter acronym
LD	Linear dichroism

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